

Enhanced superconductivity in CuH₂ monolayers

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Achieving superconductivity in hydrides at lower pressures is a long-standing scientific challenge. Here, we propose that reducing their effective dimensionality might be a possible route to achieve this goal. First-principles structural search calculations identify several two-dimensional transition metal hydrides (TMHs) with phonon-mediated superconductivity. Among them, the two CuH₂ phases with *P*-3*m*1 and *P*-6*m*2 symmetries are predicted to have the highest *T_c* values up to 44.4 and 47.8 K, comparable to the well-known MgB₂ superconductor. Their superconductivity mainly originates from the coupling of Cu 3*d_{xz}*/*d_{yz}* and H-1*s* electrons with in-plane H-vibrational phonons. Interestingly, both CuH₂ structures show a unique superconducting energy gap by solving the anisotropic Eliashberg equations. Furthermore, the superconductivity of TMHs is closely related to the polarity of the TM-H bond. Our paper paves the way for finding hydride superconductors in low-dimensional materials.

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I. INTRODUCTION

Solid hydrogen is the ideal candidate to be a high-temperature superconductor. However, its superconducting transition pressure is extremely high (i.e., above 400 GPa) [1,2] and extensive efforts have been made to find effective ways to reduce it. One of the most successful ones is the “chemical precompression” proposed by Ashcroft [3]: The reaction of hydrogen with other elements raises the electronic valence density which reduces the metallization pressure [3]. Following this principle, a large number of H-based superconductors with *T_c* values above 200 K have been predicted [4–7], and some have been successfully verified in the laboratory at pressures above 150 GPa [8–10]. Unfortunately, such pressures are still difficult to achieve. Therefore, the next challenge is to obtain high-temperature superconducting hydrides at significantly lower or even ambient pressure [11].

Like pressure as an extreme condition, the reduction in dimensionality of compounds can induce the presence of many unexpected properties that are distinct from their bulk counterparts, which mainly originate from quantum confinement and surface effects [12]. Two-dimensional (2D) materials in which all atoms are confined to a plane are regarded as the

limit expression of the atomic thickness of materials. Among the various 2D materials with exotic properties, 2D superconductors are of great interest not only for their application in constructing nanosuperconducting devices [13,14], but also as a new platform that extends the knowledge of superconductivity [14]. Until now, only several 2D superconducting materials with *T_c* values higher than 20 K have been found, including MnB₃ [15], W₂N₃ [16–18], LiBC [19], MgB₂ [20], buckled triangle borophene [21,22], and B₂MgB₂ [23]. Therefore, expanding the family of 2D superconducting materials and exploring the possibility of raising their *T_c*s are key areas of research in condensed-matter physics.

Recent synthesis of many unexpected 2D compounds, including BH [24], Ca₂N [25], Cu₈B₁₄ [26], and NaCl [27] has greatly expanded the scope of the 2D material research. On the other hand, transition-metal (TM) atoms generally act as precompressors and electron donors at high pressures, playing a vital role in stabilizing hydrides and achieving high-temperature superconductivity [6,28–30]. Furthermore, multiple *d*-orbital electronic configurations of TM atoms have demonstrated their ability to fine-tune the structures and electronic properties of their compounds. Considering the capabilities of TM and the role of reducing dimensionality, the expected superconductivity of 2D TM hydrides (TMHs) is likely to be induced.

Here, we perform a systematic search for 2D TMHs with various chemical compositions, TM_{*x*}H_{*y*} (*x*, *y* = 1–3), based

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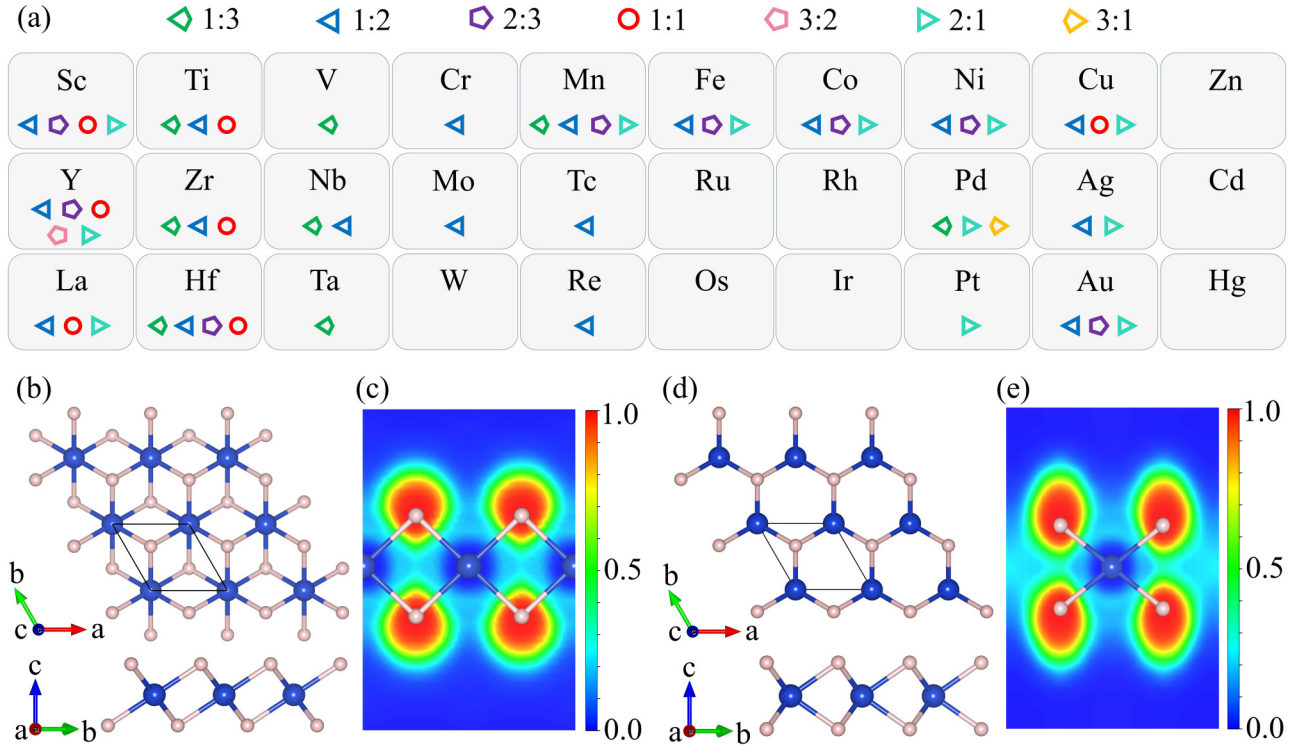


FIG. 1. (a) The distribution of stable 2D TMHs with different stoichiometries in the periodic table. For clarity, different stoichiometries are represented by different symbols. Crystal structures of (b) *t*-CuH₂ and (d) *h*-CuH₂ monolayers. ELF maps of the (c) *t*-CuH₂ in (0,1,6.3663), and (e) *h*-CuH₂ monolayers in (1,-2,0) and (2,-1,0) planes.

on first-principles structural search calculations [31]. Several identified low-energy structures show distinct electronic properties, including semiconductors, metals, superconductors, magnetic, and electrides. In this article, we focus on 2D superconducting TMHs. Among them, two CuH₂ polymorphs show the highest intrinsic T_c s of 44.4 and 47.8 K, which can be attributed to a combination of a weak charge transfer induced by Cu-H covalence and a strong coupling of in-plane H-vibrational phonons with Cu $3d_{xz}/d_{yz}$ and H-1s electrons. Our paper not only enriches H-related materials, but also stimulates an increased research on 2D superconducting materials.

II. COMPUTATIONAL METHODS

Using the particle-swarm optimization algorithm as implemented in the CALYPSO code [31,32], we have examined a number of 2D TMH structures. The vacuum distance was set to more than 15 Å to eliminate the interlayer interaction. The structural optimization and property calculations of these hydrides were completed within the DFT framework in the Vienna *ab initio* simulation package [33], adopting the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional under the generalized gradient approximation [34,35]. The phonon spectra and the electron-phonon coupling (EPC) interaction of TMH monolayers were calculated with the QUANTUM-ESPRESSO code [36]. Ultrasoft pseudopotentials were selected with a kinetic cutoff energy of 80 Ry. A $36 \times 36 \times 1$ *k*-point mesh and $12 \times 12 \times 1$ *q*-point in the first Brillouin zone were used in the EPC calculations

for TMHs. T_c values were calculated using the Allen and Dynes modified McMillan equation [37]. Anisotropic superconducting properties were investigated by solving the fully anisotropic Migdal-Eliashberg equations as implemented in the electron-phonon Wannier (EPW) code [38–40]. Detailed structural prediction and adopted parameters can be found in the Supplemental Material [41].

III. RESULTS AND DISCUSSION

Our extensive structural prediction yields 73 stable TMH monolayers, whose chemical compositions and the corresponding TM atoms can be found in Fig. 1(a). In addition to the structures reported by the Guo [49] and Zeng [50] groups, we also found a series of new TMHs (Fig. S1 in the Supplemental Material [41]). Additionally, they have negative cohesive energies and diverse electronic attributes (Fig. S2 in the Supplemental Material [41]). A large number of 2D TMHs exhibit metallicity, but only eight are superconductors (Fig. S3 in the Supplemental Material [41]). Subsequently, we mainly discuss the two polymorphs of CuH₂ with the highest T_c values. One CuH₂ with *P*-3 m 1 symmetry is identical to 1T-MoS₂ [referred to as *t*-CuH₂ hereinafter, Fig. 1(b)], and the other one with *P*-6 m 2 symmetry is designated as *h*-CuH₂ [Fig. 1(d)] due to similar spatial configurations to 2H-MoS₂ [51]. Based on the calculated cohesive energy, *t*-CuH₂ is more stable than *h*-CuH₂ (Table S1 in the Supplemental Material [41]). The Cu-H bond lengths of *t*- and *h*-CuH₂ are 1.77 and 1.79 Å, respectively, which is comparable to 1.73 Å of Cu-H in CuH [52]. The electron localization function (ELF)

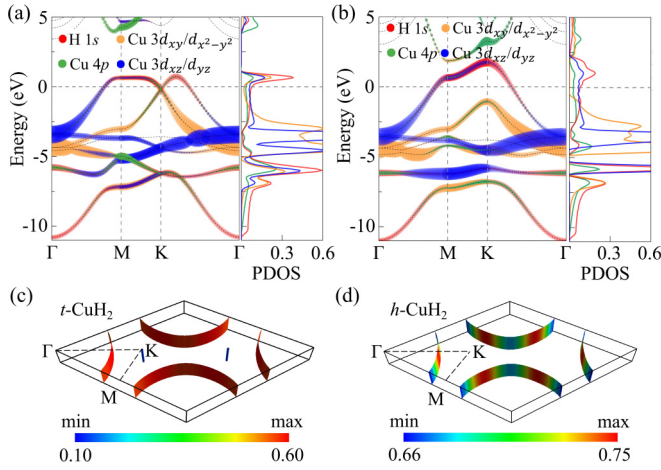


FIG. 2. The projected electronic band structures and PDOS of (a) $t\text{-CuH}_2$ and (b) $h\text{-CuH}_2$ calculated at the PBE level and their corresponding FLs in (c) and (d), respectively.

maps display a ionic interaction between Cu and H atoms [Figs. 1(c) and 1(e)], which is also supported by the Bader charge analysis [53] (e.g., a H atom gains $0.23e^-$ from a Cu atom). On the other hand, the similar electronegativity of Cu and H atoms (1.9 vs 2.1) and their strong orbital overlap in CuH_2 compounds indicate a measure of the covalency in the Cu-H bond [Figs. 2(a) and 2(b)]. Therefore, the nature of the Cu-H bond can be characterized as a mixture of a ionic and covalent bond as observed in LaH_{10} [54]. Remarkably, the negatively charged H atoms in CuH_2 compounds show some analogies with those in clathrate hydrides, which is in contrast to the strongly covalent atomic hydrogen in H_3S [55,56].

The projected electronic bands, projected density of states (PDOS), and Fermi lines (FLs) of $t\text{-}$ and $h\text{-CuH}_2$ [Fig. 2] are calculated to investigate their electronic properties. For $t\text{-CuH}_2$, one band crosses the Fermi level and forms two Fermi lines: a large circle near the Brillouin zone (BZ) boundary is associated with the hybridization of Cu $3d_{xz}/d_{yz}$ and H-1s states, and a small cylindrical electron pocket centered at the K point is derived from Cu $3d_{xy}/d_{x^2-y^2}$, $3d_{xz}/d_{yz}$, $4p$, and H-1s states [Figs. 2(a) and 2(c)]. The maximum Fermi velocity is located at the big circular FL, whereas the minimum one is found at the small cylindrical electron pocket. More interestingly, a Dirac cone (DC) appears at the K point near the Fermi level, originated by the hybridization between Cu $3d_{xy}/d_{x^2-y^2}$, $3d_{xz}/d_{yz}$, $4p$, and H-1s orbitals. The $h\text{-CuH}_2$ only has one FL around the BZ boundary, associated with the hybridization between Cu $3d_{xz}/d_{yz}$, and H-1s orbitals [Figs. 2(b) and 2(d)], which contrasts sharply with the presence of in-plane Cu $3d_{xy}/d_{x^2-y^2}$, Cu- $4p$ states, and a DC near the Fermi level in $t\text{-CuH}_2$. The Fermi velocity maximum is located at the FL along the Γ - K direction. These differences near the Fermi level in the metallic character of $t\text{-CuH}_2$ and $h\text{-CuH}_2$ may originate from the distinct H-ligand fields around Cu atoms, namely, octahedral coordination for $t\text{-CuH}_2$ and triangular-prism coordination for $h\text{-CuH}_2$. It is well known that ligand atoms in a trigonal prism field have a larger repulsion interaction than in an octahedral field, thus, leading to a strong splitting of Cu-3d orbitals and a broaden dispersion

of electronic bands in $h\text{-CuH}_2$. On the other hand, the small splitting in the octahedral field induces a strong interaction between Cu-3d orbitals, resulting in the formation of a DC at the K point in $t\text{-CuH}_2$ [57]. Significant van Hove singularities appear near the Fermi level for $t\text{-CuH}_2$ and $h\text{-CuH}_2$ in the PDOS, which may induce a strong electron-phonon coupling. Since $t\text{-CuH}_2$ has a large band gap above 0.7 eV, we recalculated the electronic-band structure (Fig. S5 in the Supplemental Material [41]) to verify its metallicity using the HSE06 functional [58]. In addition to reproducing its metallic character, the DC at the K point below the Fermi level also appears.

Next, we explore the superconductivity of $t\text{-}$ and $h\text{-CuH}_2$ in Figs. 3(a)–3(d). The absence of imaginary phonon modes indicates that $t\text{-}$ and $h\text{-CuH}_2$ are dynamically stable. Obviously, the PhDOS can be separated into two regions, the vibration modes of Cu atoms dominate low-frequency acoustic branches, whereas the vibrations of H atoms are responsible for the middle- and high-frequency optical branches. The Eliashberg spectral function $\alpha^2F(\omega)$ of $t\text{-CuH}_2$ has a couple of large peaks in the region of $356.08\text{--}1036.04\text{ cm}^{-1}$, corresponding to soft modes along Γ - M and K - Γ directions and a rather flat phonon dispersion along the M - K direction, which mainly originates from the in-plane E_g [twofold degenerate Raman active modes in Fig. 3(a)] and E_u (twofold degenerate infrared active modes in Figs. S6(c) and S6(d) in the Supplemental Material [41]) modes mainly associated with H atoms. Remarkably, the contributions to EPC (λ) of E_g and E_u modes represent 62% of the total value where the E_g mode in $t\text{-CuH}_2$ shows a stronger EPC and dominates the Eliashberg function. However, the soft modes in the acoustic branches along the Γ - M and K - Γ paths contribute a non-negligible 33% of the total λ . Combining it with the electronic band structure, we can conclude that superconductivity in $t\text{-CuH}_2$ mainly comes from the E_g mode of H atoms strongly coupled to Cu $3d_{xy}/d_{x^2-y^2}$, $3d_{xz}/d_{yz}$, $4p$, and H-1s states, similar to HfH_{10} [30]. The resulting λ of $t\text{-CuH}_2$ is 1.20, larger than bulk MgB_2 (0.77) [59]. Based on the Allen-Dynes modified McMillan equation [37], the T_c of $t\text{-CuH}_2$ is estimated to be 44.4 K with $\mu^* = 0.1$.

Likewise, the medium frequency modes E' and E'' (associated with the displacement of H atoms in Figs. 3(c), S7(c) and S7(d) in the Supplemental Material [41]) contribute to more than half (51%) of λ in $h\text{-CuH}_2$ where the contribution from E' (twofold degenerate H in-plane mode) is very essential, as was also found in SnH_4 and H_4Te at high pressure [60,61]. Notably, the three acoustic branches contribute to 45% to the total λ , which is due to the soft modes along Γ - M , M - K , and K - Γ [Fig. 3(d)]. The calculated EPC constant, $\lambda = 1.42$, is slightly higher than that of $t\text{-CuH}_2$, leading to a T_c of 47.8 K, which is even higher than 39 K in MgB_2 , consisting of graphenelike B layers, a typical representative of traditional superconductors [62]. On the other hand, both CuH_2 and MgB_2 show a common feature of λ dominated by in-plane vibration modes [62,63]. To explore the origin of the high- T_c superconductivity in CuH_2 , we explored six TMHs with the same $h\text{-CuH}_2$ structure (Fig. S8 in the Supplemental Material [41]). For the same periodic TM atoms (e.g., V vs Cu, Nb vs Mo, and Ta vs W), the lower the transferred charge, the higher the T_c [Fig. 3(e)]. Among them, Cu transfers the lowest

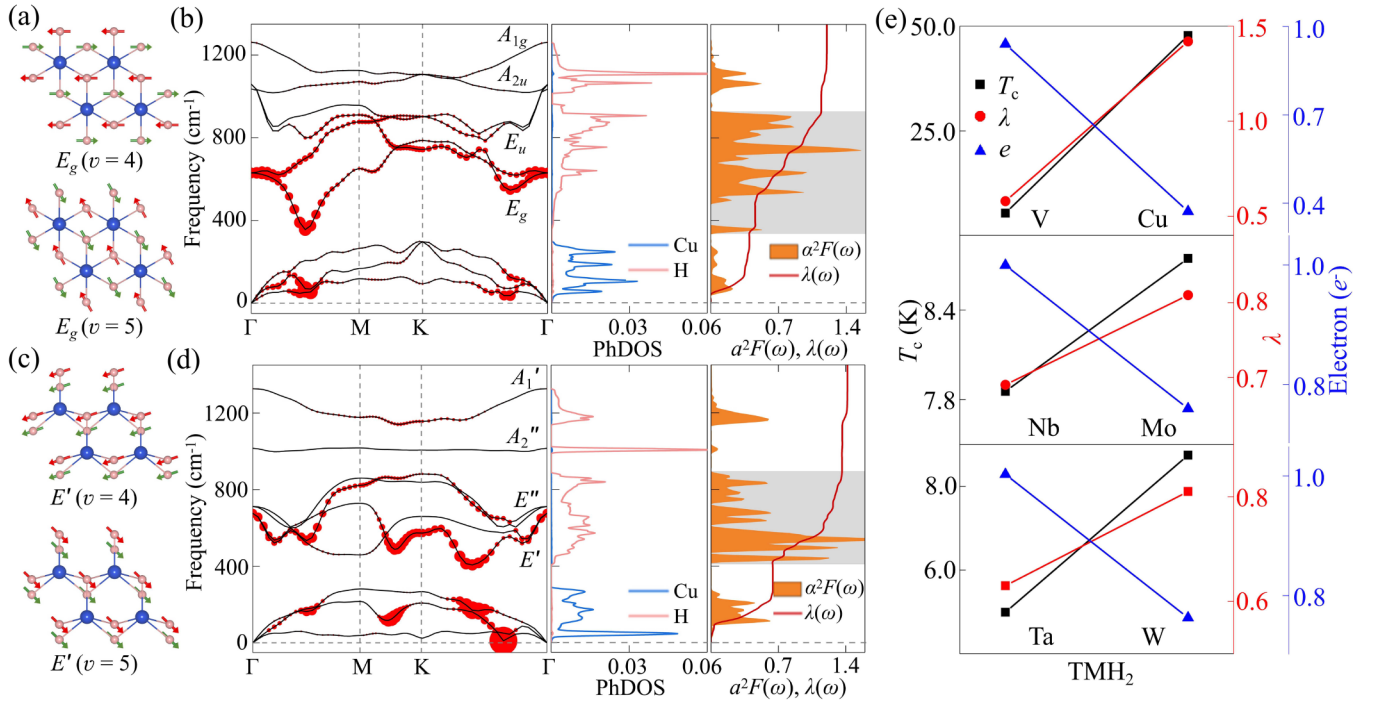


FIG. 3. The two degenerate E_g modes in (a) t -CuH₂ and the two degenerate E' modes in (c) h -CuH₂. The phonon dispersion curve PhDOSs, Eliashberg spectral function $\alpha^2F(\omega)$, and integrated EPC consistent λ for (b) t -CuH₂ and (d) h -CuH₂. (e) The variation of T_c , λ , and the amount of transferred electrons for six TMH₂ superconductors with $P-6m2$ symmetry.

charge. These results indicate that a greater covalent character of TM-H could be a factor favoring superconductivity.

Considering that the EPW can better describe the superconducting energy gap and the EPC of layered or low-dimensional systems with respect to the traditional McMillan approximation [39], we, therefore, apply the fully anisotropic Migdal-Eliashberg equations to explore the superconducting energy gap Δ_{nk} and the EPC parameter λ_{nk} on the FL with $\mu_c^* = 0.13$ as previous works [16–20]. At 5 K t -CuH₂ shows a single superconducting energy gap since the normalized superconducting density of states (SDOS) has a peak at 10.3 meV [Fig. 4(b)]. This is reasonable because t -CuH₂ has one band that crosses the Fermi level to form two FLs [Fig. 2(c)], which do not present significantly different orbital contributions so that do not lead to the formation of multiple energy gaps. This feature is distinct from the multigap MgB₂ [64,65] and LiBC [19] monolayers where the FLs result from σ and π orbitals, leading to separate superconducting gaps. The largest EPC λ_{nk} are associated with Cu $3d_{xy}/d_{x^2-y^2}$, $3d_{xz}/d_{yz}$, $4p$, and H- $1s$ states at the K point and along Γ - M with a low Fermi velocity. The resulting λ_{nk} , up to 2.4, is about twice as large as the isotropic one reported above. At $T = 5$ K, the Δ_{nk} is located at the K point and in the Γ - M direction on the FL with a maximum value of 10.3 meV, showing distributions similar to that of the EPC. The superconducting energy gap vanishes at 64.3 K, which is identified as the critical temperature T_c [Fig. 4(a)]. This is slightly lower than 65 K of LiBC with $\mu_c^* = 0.13$ [19]. The h -CuH₂ is also a single energy gap superconductor with T_c value close to 72.5 K (Fig. S10 in the Supplemental Material [41]). The largest superconducting energy gap and EPC are also in the Γ - M direction with Δ_{nk} and λ_{nk} equal to 14.2 meV and 4.9 at $T = 5$ K, respectively.

Our paper has shown that 2D hydrogen-based materials might help achieve hydride superconductivity at ambient pressure. As in bulk hydride superconductors, their superconductivity depends not only on the constituent elements, but also on their chemical compositions [28] so that T_c values of 2D hydrides have a large range of improvement. On the other hand, just like superconductivity in 2D hydrides, there could also be a strong possibility of achieving it in one- and zero-dimensional hydrides. It is believed that our paper will

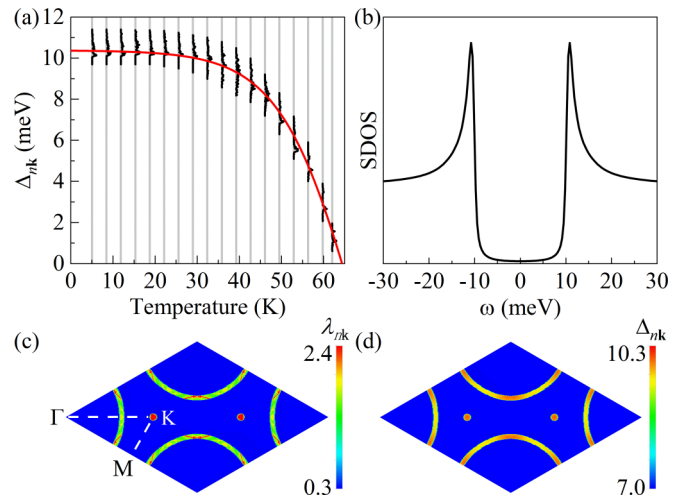


FIG. 4. (a) Energy distribution of superconducting gaps Δ_{nk} versus T and (b) SDOS at $T = 5$ K for the t -CuH₂ monolayer. The momentum-resolved EPC (c) λ_{nk} and superconducting gaps (d) Δ_{nk} on the FL at $T = 5$ K.

stimulate further research on low-dimensional hydride superconductors.

IV. CONCLUSIONS

To summarize, our comprehensive first-principles structural searches for 2D TMHs have identified 73 structures displaying diverse electronic properties, such as superconductivity, magnetism, and electrides. Their negative cohesive energies and high dynamical stabilities demonstrate their synthetic feasibilities. Focusing on superconductivity, two CuH₂ phases are predicted to have a remarkably high T_c of 44.4 and 47.8 K. The medium-frequency and in-plane H-atom vibrational modes are mainly responsible for the EPC. For 2D TMHs, their superconductivity depends on the polarity of the TM-H bonds. Our results represent an important breakthrough in the understanding of hydrogen-based su-

perconductors and will inevitably drive further experimental exploration.

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